ECON 5345 Homework 6 Report

Harlly Zhou

March 19, 2026

Question 1

a. Since Ω has a Cholesky decompositive, it is of full rank, denoted by n. Then

$$\text{rank}(\Omega) = \text{rank}(C'C) = \text{rank}(C) = n.$$

Since C is full rank, it is invertible.

b. Taking variance of e_t gives

$$\begin{aligned} CC' = \Omega = \text{Var}(e_t) &= \text{Var}(D(0)v_t) \\ &= D(0)\,\text{Var}(v_t)D(0)' \\ &= D(0)D(0)'. \end{aligned}$$

Therefore, $R = C^{-1}D(0) = C'[D(0)']^{-1}$. It follows that

$$RR' = C'[D(0)']^{-1}D(0)(C')^{-1} = I.$$

c. It is immediate from simple linear algebra:

$$R_O(\theta)\begin{bmatrix}1 & 0 \\ 0 & -1\end{bmatrix} = \begin{bmatrix}\cos(\theta) & -(-\sin(\theta)) \\ \sin(\theta) & -\cos(\theta)\end{bmatrix} = R_E(\theta).$$

Define $J = \begin{bmatrix}1 & 0 \\ 0 & -1\end{bmatrix}$. Note that J only flips the sign of the second shock. Suppose $R = R_E(\theta)$. Then we have

$$e_t = CR_E(\theta)v_t = CR_O(\theta)J^{-1}v_t.$$

We can redefine the strutural shock to be $J^{-1}v_t$, and that is WLOG since the identification is up to signs.